Part VIII — Electricity and Magnetism · Chapter 44

Magnetism

How moving charge creates magnetic fields and how those fields exert forces

> *A compass needle aligns with Earth's field. An MRI machine threads magnetic flux through the human body to image soft tissue. A maglev train floats centimetres above its track, propelled and levitated purely by magnetic forces. A speaker converts oscillating electrical current into vibrating air. Iron filings arrange themselves around a bar magnet in graceful arcs. All of these phenomena arise from a single underlying reality: magnetic fields are created by moving electric charges, and they exert forces on other moving charges. Magnetism is not a separate force from electricity — it is electricity in motion. This chapter introduces the magnetic field, magnetic forces, and the close relationship between electricity and magnetism that Einstein would later show to be a consequence of special relativity.*

44.1 What Magnetism Is

Magnetism is an interaction between moving electric charges. Every magnetic effect — from the attraction between a magnet and a refrigerator door to the operation of a particle accelerator — can be traced ultimately to electric charge in motion.

This connection was not understood until the 19th century. Ancient peoples knew that lodestones (naturally magnetic iron ore) attracted iron and that compass needles aligned with a directional force. For millennia, magnetism was treated as a phenomenon entirely separate from electricity. This changed in 1820, when Hans Christian Ørsted discovered that an electric current deflected a nearby compass needle — proving for the first time that electricity and magnetism are related. The decades that followed produced a complete mathematical framework (Maxwell's equations) unifying the two.

At the atomic level, magnetism arises from two sources: the orbital motion of electrons around atomic nuclei (orbital magnetic moment) and the intrinsic spin of electrons (spin magnetic moment). The magnetic properties of bulk materials emerge from the combined effects of these atomic-scale magnetic moments.

Historical Note: Ørsted's discovery in 1820 triggered a rapid succession of breakthroughs: Ampère quantified the force between current-carrying wires (1820–1825), Faraday discovered electromagnetic induction (1831), and Maxwell synthesised all electromagnetic phenomena into four equations (1865). This 45-year sprint from a compass deflection to a complete theory of light as an electromagnetic wave stands as one of the greatest intellectual achievements in the history of physics.

44.2 Magnetic Poles

Every magnet has two regions called poles — the north pole (N) and south pole (S) — where the magnetic force is strongest. Poles always come in pairs: a magnetic monopole (an isolated north or south pole) has never been observed in nature, despite being theoretically interesting.

The law of magnetic poles parallels the law of electric charges: like poles repel each other; unlike poles attract. Two north poles brought together push apart. A north and south pole attract. This similarity to electric charge is not coincidental — it reflects the deep connection between the two phenomena.

However, unlike electric charge (which can be separated — you can have a positive object and a negative object independently), magnetic poles cannot be isolated. Break a bar magnet in half and you get two smaller bar magnets, each with a north and south pole. Continue dividing and you continue getting complete magnets. All the way down to individual atoms, magnetism is dipolar.

Key Point: There is no magnetic monopole: no isolated north or south pole. Every magnet, regardless of size, has both a north and a south pole. This is expressed in Maxwell's equations by the law $\nabla \cdot B = 0$ (magnetic field lines have no sources or sinks — they form closed loops).

44.3 Attraction, Repulsion, and Magnetic Domains

In ferromagnetic materials (iron, nickel, cobalt, and their alloys), atoms group into microscopic regions called magnetic domains. Within each domain, all atomic magnetic moments are aligned, producing a strong net magnetic moment for the domain. The domain itself acts like a tiny bar magnet.

In an unmagnetised piece of iron, the domains point in random directions, so their magnetic effects cancel. When an external magnetic field is applied, domains whose magnetic moments align with the field grow (their boundaries shift) at the expense of misaligned domains. If the external field is strong enough, all domains align and the material becomes strongly magnetised — this is the process of magnetisation.

In permanent magnets (such as those made from alnico or neodymium-iron-boron alloys), the domains remain aligned even after the external field is removed. In soft magnetic materials (such as annealed iron), the domains readily realign with a new applied field, but return to random orientation when the field is removed — making them suitable for electromagnets and transformer cores.

Heating a magnet above a material-specific temperature called the Curie temperature disrupts the domain structure and destroys the permanent magnetism. The Curie temperature for iron is 770°C. A magnet dropped, shocked, or strongly vibrated also loses some magnetisation as the impact scrambles the domains.

> **Example — Domain explanation of attraction** When a magnet is brought near an unmagnetised iron nail, the magnet's field causes the domains in the nail to partially align. The aligned nail then has its own north and south poles, with the pole closest to the magnet being the opposite polarity. Opposite poles attract — the nail is drawn to the magnet. This induced magnetisation explains why magnets attract both poles of an iron object.

44.4 Magnetic Fields and Field Lines

A magnetic field (B) is a region of space in which a magnetic force can be experienced by a moving charge or a magnetic pole. The magnetic field is a vector quantity — it has both magnitude and direction at every point.

The SI unit of magnetic field strength is the tesla (T), named for Nikola Tesla. 1 T is a very strong field in everyday terms — a strong bar magnet produces about 0.1 T near its poles; a refrigerator magnet about 10 mT; Earth's field about 50 μT; an MRI machine 1.5–3 T.

Magnetic field lines are a visual tool for representing the direction and strength of a magnetic field. The rules for field lines are: (1) Field lines emerge from north poles and enter south poles in the region outside the magnet. Inside the magnet, field lines run from south to north, making them continuous closed loops. (2) The direction of the field line at any point gives the direction of B at that point (the direction in which a north pole would be pushed). (3) The density of field lines (lines per unit area perpendicular to the lines) represents the magnitude of B. More closely spaced lines indicate stronger fields.

Unlike electric field lines, which begin at positive charges and end at negative charges, magnetic field lines are always closed loops — they have no starting or ending points. This is a fundamental difference between electric and magnetic fields.

Source	Approximate B (T)
Earth's surface (average)	5×10^{-5} T (50 μT)
Refrigerator magnet	~5×10^{-3} T (5 mT)
Bar magnet (near pole)	~0.1 T
Loudspeaker magnet (air gap)	~1 T
Clinical MRI machine	1.5–3 T
Research MRI (record)	~21 T
Neutron star surface	~10^{8} T
Magnetar (extreme neutron star)	~10^{11} T

Table 44.1 *Representative magnetic field strengths in SI units.*

44.5 Earth's Magnetic Field

Earth behaves like a giant magnet with a magnetic field that extends from the interior out into space (the magnetosphere). The geomagnetic north pole (which attracts compass north poles) is actually a magnetic south pole in the physical sense — it attracts north poles of other magnets, so by the pole convention it is a south pole of Earth's magnetic dipole. Earth's geographic north pole and magnetic north pole are close but not coincident; the angular difference is called magnetic declination, which varies by location and changes slowly over time.

Earth's field is generated by convective motion of molten iron in the outer core — a mechanism called the geodynamo. Detailed models are complex, but the basic principle is that electrical currents circulating in the conductive liquid iron produce magnetic fields (by Ampère's law, covered in section 44.8). The field has not been constant over geological time: it has reversed polarity hundreds of times, with reversals typically taking thousands of years.

The magnetosphere deflects the solar wind (streams of charged particles from the Sun) away from Earth's surface. Without this magnetic shield, the solar wind would strip away the atmosphere over geological timescales, as it did for Mars after Mars lost its protective magnetic field.

The practical consequence of Earth's field for navigation is magnetic declination. In much of North America, a compass points several degrees west of geographic north. Navigators and hikers must correct for declination when using a magnetic compass to determine true north.

44.6 Magnetic Force on Moving Charges

A charged particle moving through a magnetic field experiences a force called the Lorentz force (or magnetic force). The magnitude of the force depends on the charge, the speed, the field strength, and the angle between the velocity and the field.

$$F = qvB \sin\theta$$

Magnetic force on a moving charge. F = force (N); q = charge (C); v = speed (m/s); B = magnetic field (T); θ = angle between velocity v and field B. Maximum force ($F = qvB$) occurs when $v \perp B$ ($\theta = 90°$). Zero force when $v \parallel B$ ($\theta = 0$).

The direction of the magnetic force is always perpendicular to both the velocity v and the field B. This is determined by the right-hand rule: point the fingers of the right hand in the direction of v, curl them toward B; the thumb points in the direction of F for a positive charge. For a negative charge, the force is in the opposite direction.

Because the magnetic force is always perpendicular to v, it never does work on the particle ($W = F \cdot d$, and $F \perp d$ when $F \perp v$). The magnetic force changes the direction of motion but not the speed or kinetic energy. This is why charged particles moving in a uniform magnetic field travel in circles (or helices if there is a component of velocity along B): the force acts as a centripetal force that continuously redirects v without changing its magnitude.

$$r = mv / qB$$

Radius of circular motion in a magnetic field. r = radius (m); m = particle mass (kg); v = speed (m/s); q = charge (C); B = field (T). Larger mass, higher speed, or weaker field → larger radius. Larger charge or stronger field → smaller radius.

Example — Circular motion in B A proton ($m = 1.67 \times 10^{-27}$ kg, $q = 1.6 \times 10^{-19}$ C) moves at 3.0×10^6 m/s perpendicular to a 0.25 T field. $F = qvB = (1.6 \times 10^{-19})(3.0 \times 10^6)(0.25) = 1.2 \times 10^{-13}$ N. Radius $r = mv/qB = (1.67 \times 10^{-27})(3.0 \times 10^6)/[(1.6 \times 10^{-19})(0.25)] = 0.125$ m = 12.5 cm.

Key Point: The magnetic force can change the direction of a charged particle but never its speed or kinetic energy. This is because the force is always perpendicular to the velocity. The particle curves, but it does not speed up or slow down.

44.7 Magnetic Force on Current-Carrying Wires

Since electric current consists of moving charges, a current-carrying wire in a magnetic field experiences a magnetic force. The force on a straight wire of length L carrying current I in a field B at angle θ to the wire is:

$$F = BIL \sin\theta$$

Magnetic force on a current-carrying wire. F = force (N); B = magnetic field (T); I = current (A); L = length of wire in the field (m); θ = angle between wire direction and B. Maximum force when wire is perpendicular to B (θ = 90°).

The direction of the force is given by the right-hand rule applied to conventional current: point the right-hand fingers in the direction of conventional current flow, curl them toward B; the thumb gives the direction of F.

This force on current-carrying wires is the operating principle of electric motors and loudspeakers. In a motor, the wire carries current in a magnetic field; the force rotates the wire. In a loudspeaker, an oscillating current in a voice coil (in the field of a permanent magnet) produces an oscillating force that vibrates the cone, producing sound.

Example — Force on wire A 0.30 m wire carries 4.0 A of current perpendicular to a 0.080 T magnetic field. F = BIL sin 90° = (0.080)(4.0)(0.30)(1) = 0.096 N. If the current is reversed, the force reverses direction.

44.8 Magnetic Fields Around Currents

Just as moving charges experience magnetic forces, moving charges also create magnetic fields. A current-carrying wire is surrounded by a magnetic field that forms concentric circles centred on the wire. The direction is given by the right-hand rule for the source: wrap the right hand around the wire with the thumb pointing in the direction of conventional current; the fingers curl in the direction of the circular magnetic field lines.

The magnitude of the magnetic field at distance r from a long straight wire carrying current I is:

$$B = \mu_0 I / (2\pi r)$$

Magnetic field of a long straight wire. $\mu_0 = 4\pi \times 10^{-7}$ T·m/A is the permeability of free space; I = current (A); r = perpendicular distance from wire (m). Field decreases as 1/r from the wire.

Two parallel wires carrying current in the same direction attract each other; in opposite directions, they repel. This force between wires forms the basis of the historical definition of the ampere: 1 A was defined as the current in each of two parallel wires 1 m apart that produces a force of 2×10^{-7} N per metre of length between them. (The 2019 SI revision redefined the ampere in terms of the elementary charge, but the physics is unchanged.)

> **Example — Field near a wire** A wire carries I = 5.0 A. Find B at r = 2.0 cm = 0.020 m. $B = (4\pi \times 10^{-7})(5.0)/[2\pi(0.020)] = (2 \times 10^{-6})(5.0)/0.020 = 5.0 \times 10^{-5}$ T = 50 μT — comparable to Earth's field, which is why power lines affect sensitive compasses nearby.

44.9 Electromagnets

An electromagnet is a device that produces a strong magnetic field only when current flows through it. It consists of a coil of wire (a solenoid) wrapped around a ferromagnetic core (usually iron). The current creates a magnetic field; the iron core concentrates and amplifies the field enormously by aligning its magnetic domains.

The magnetic field inside a solenoid (a long, tightly wound coil) without a core is:

$$B = \mu_0 n I$$

Magnetic field inside a solenoid. $\mu_0 = 4\pi \times 10^{-7}$ T·m/A; n = number of turns per unit length (turns/m); I = current (A). With a ferromagnetic core, B is multiplied by the relative permeability μ_r, which can be hundreds to tens of thousands for iron.

Electromagnets have three key advantages over permanent magnets: (1) the field can be switched on and off instantly by switching the current, (2) the field strength is controllable by adjusting the current, and (3) the polarity can be reversed by reversing the current. These properties make electromagnets essential for electric motors, generators, relays, MRI machines, industrial lifting magnets, and particle accelerators.

A relay is an electrically operated switch: a small current through an electromagnet creates a field that pulls a switch closed (or opens it), allowing the small current to control a much larger current in a separate circuit. Relays are the ancestors of transistors and were used in early computers and telephone exchange systems.

44.10 Motors and Magnetic Force

An electric motor converts electrical energy to mechanical (rotational) energy using the force on a current-carrying conductor in a magnetic field.

In a simple DC motor, a rectangular coil of wire (the armature) is placed between the poles of a permanent magnet. When current flows through the coil, the two sides perpendicular to the magnetic field experience forces in opposite directions (by $F = BIL$), producing a torque that rotates the coil. As the coil rotates, a device called a commutator reverses the current direction every half-turn, keeping the torque in the same rotational direction so the coil continues spinning.

Real motors use multiple coils and pole pairs to produce smooth, continuous torque. The efficiency of modern electric motors (85–98%) is much higher than internal combustion engines (20–40%), which is one reason electric vehicles have a significant energy advantage per kilometre travelled.

A generator is the reverse of a motor: mechanical rotation of a coil in a magnetic field induces a voltage in the coil (electromagnetic induction, Chapter 45). Motors and generators use identical physical principles in opposite directions. The same machine can function as both: electric vehicle regenerative braking works by running the motor as a generator.

44.11 Magnetic Materials

Not all materials respond to magnetic fields in the same way. There are three main categories based on how the material interacts with an external magnetic field.

Type	Behaviour in External B	Examples	Cause
Ferromagnetic	Strongly attracted; can be permanently magnetised	Iron, steel, nickel, cobalt, neodymium alloys	Large magnetic domains that align with and amplify B
Paramagnetic	Weakly attracted; not permanently magnetised	Aluminium, platinum, oxygen gas	Unpaired electron spins partially align with B; thermal motion destroys alignment
Diamagnetic	Weakly repelled; creates opposing internal field	Copper, water, bismuth, most organic compounds	Electron orbits slightly distorted by B, creating opposing moment; effect very small

Table 44.2 *Classification of magnetic materials. Ferromagnetic materials are the only practically significant category for everyday magnetism.*

Superconductors (Chapter 50) are perfect diamagnets below their critical temperature: they expel all magnetic flux from their interior (Meissner effect). This is what allows a magnet to levitate above a superconductor — a dramatic demonstration of perfect diamagnetism. Maglev trains in Japan use a different approach (active electromagnets with controlled feedback) rather than superconducting diamagnetism directly, though research maglev systems do use superconducting magnets.

44.12 Everyday and Engineering Applications of Magnetism

Magnetism pervades modern technology. The following survey illustrates how the principles of this chapter appear in widely used systems.

Speakers and Microphones

A loudspeaker uses the force on a voice coil (carrying audio-frequency current) in the field of a permanent magnet to vibrate a paper or polymer cone. The vibrating cone pushes air, producing sound. A microphone is the reverse: a diaphragm vibrated by sound moves a coil in a magnetic field, inducing a voltage (by Faraday's law, Chapter 45) that represents the sound electrically.

Magnetic Storage (Hard Drives, Magnetic Tape)

Traditional hard drives store data by magnetising tiny regions of a rotating disk in one of two orientations (representing 0 and 1). A read/write head containing a small electromagnet writes data by magnetising regions and reads data by detecting the flux change from previously magnetised regions as they pass beneath the head. While SSDs (solid-state drives) are replacing hard drives in many applications, the technology illustrates the controlled use of domain orientation for information storage.

Medical Imaging (MRI)

Magnetic resonance imaging uses a very strong magnetic field (1.5–3 T, produced by superconducting electromagnets) to align hydrogen nuclei in the body. Radio-frequency pulses tip the nuclei out of alignment; as they relax back, they emit radio signals that are detected and used to reconstruct images of soft tissue with superb contrast. MRI avoids ionising radiation (unlike X-rays or CT scans), making it preferred for soft-tissue imaging.

Electric Power Generation and Transmission

Virtually all electrical power is generated by rotating coils in magnetic fields (electromagnetic induction). Coal, gas, nuclear, hydroelectric, and wind generators all spin a shaft that drives a rotating armature through a magnetic field. The resulting alternating voltage is transformed up to high voltage (100–750 kV) for transmission and then stepped down at local substations using transformers — devices that also rely on magnetic flux linkage.

Particle Accelerators

High-energy physics experiments use powerful electromagnets to steer charged particle beams along circular paths (recall $r = mv/qB$). The Large Hadron Collider at CERN uses over 1200 superconducting dipole magnets (each producing ≈ 8 T) to keep proton beams on their 27 km circular track.

44.13 Common Mistakes in Magnetism Problems

- Confusing magnetic field direction with force direction. The magnetic force on a positive charge is $F = qv \times B$, which is perpendicular to both v and B. The force is not in the direction of B. Always apply the right-hand rule explicitly; do not guess the force direction from the field direction.
- Applying the right-hand rule to the wrong quantity. There are two right-hand rule applications: one for the force on a charge (fingers along v, curl toward B, thumb = F for positive charge) and one for the field around a wire (thumb along current, fingers curl in the direction of B). Confusing the two leads to reversed directions.
- Forgetting the sinθ factor. The magnetic force is $F = qvB \sin\theta$, not $F = qvB$. A charge moving parallel to B ($\theta = 0$) experiences zero magnetic force. Always identify the angle between v and B before computing the force magnitude.
- Thinking the magnetic force does work. Because the magnetic force is always perpendicular to v, it never does work on a moving charge. The speed of a charged particle moving in a pure magnetic field is constant; only its direction changes. A changing magnetic field can do work (via electromagnetic induction), but a static magnetic field on a moving charge cannot.
- Confusing geographic north with magnetic north (for compass problems). Earth's geographic north pole is near the magnetic south pole. Compass north (the direction the north pole of a compass needle points) is toward Earth's magnetic south pole — and the magnetic declination angle must be applied to convert compass bearing to true north bearing.
- Forgetting that magnetic field lines are closed loops. Electric field lines start on positive charges and end on negative charges. Magnetic field lines have no start or end points — they are always closed loops. Sketching field lines that start or end at a point is physically wrong.

Common Mistake: In magnetism force problems, always determine direction first using the right-hand rule, then compute magnitude using $F = qvB \sin\theta$ or $F = BIL \sin\theta$. Writing only a number without stating the direction is an incomplete answer.

Chapter Summary

Magnetism arises from moving electric charges. Magnetic fields exert forces on moving charges and current-carrying conductors. The same principles that explain bar magnets also explain electric motors, generators, and the technology of modern civilisation.

- Magnetic poles: north and south; like poles repel, unlike attract. Monopoles do not exist; field lines are closed loops.
- Ferromagnetism: magnetic domains; alignment by external field; Curie temperature; permanent vs. soft magnets.
- Magnetic field B: vector, SI unit tesla (T). Field lines run N to S outside the magnet, S to N inside.
- Magnetic force on charge: $F = qvB \sin\theta$; direction by right-hand rule; always perpendicular to v; never does work.
- Circular motion in B: $r = mv/qB$; centripetal force provided by magnetic force.
- Magnetic force on wire: $F = BIL \sin\theta$; basis of electric motors.
- Field of straight wire: $B = \mu_0 I/(2\pi r)$; field circles wire; right-hand rule for direction.
- Solenoid field: $B = \mu_0 nI$; amplified by ferromagnetic core in electromagnets.
- Applications: motors, speakers, MRI, hard drives, power generation, particle accelerators.

Key Terms

magnetic field (B): A vector field in space that exerts forces on moving charges and magnetic materials; SI unit tesla (T).

magnetic pole: Either of the two regions of a magnet (north, N, or south, S) where the field is strongest; poles always occur in pairs.

magnetic domain: Microscopic region in a ferromagnetic material in which all atomic magnetic moments are aligned; determines macroscopic magnetic properties.

ferromagnetism: Strong magnetic behaviour arising from aligned domains; found in iron, nickel, cobalt, and their alloys; can be permanently magnetised.

Curie temperature: Temperature above which a ferromagnetic material loses its permanent magnetism due to thermal disruption of domain alignment.

magnetic field line: A line whose direction at every point gives the direction of B; field lines form closed loops (no sources or sinks).

tesla (T): SI unit of magnetic field strength: 1 T = 1 N/(A·m). 1 T is a strong field; Earth's surface field is ~50 μT.

Lorentz force: The magnetic force on a moving charge: $F = qvB \sin\theta$; always perpendicular to both v and B; never does work.

right-hand rule: Rule for finding the direction of the magnetic force or field: for force ($F = qv \times B$), point fingers in direction of v, curl toward B, thumb = direction of F for positive charge; for field around wire, thumb in direction of current, fingers curl in direction of B.

solenoid: A tightly wound coil of wire; produces a nearly uniform magnetic field inside: $B = \mu_0 nI$.

electromagnet: A solenoid with a ferromagnetic core; field controlled by current; can be switched on/off and have its strength and polarity adjusted.

permeability of free space (μ_0): $\mu_0 = 4\pi \times 10^{-7}$ T·m/A; the magnetic constant relating current to the magnetic field it produces in a vacuum.

magnetic declination: The angle between geographic north and magnetic north at a given location on Earth's surface; must be applied when using a compass for true-north navigation.

Concept Review Questions

Answer in complete sentences. No calculations required unless stated.

1. Explain why breaking a bar magnet in half does not produce an isolated north pole and an isolated south pole. What does it produce, and why?
2. In your own words, explain what magnetic domains are and how they explain why iron is strongly attracted to a magnet while copper is not.
3. A charged particle moves through a uniform magnetic field without experiencing any magnetic force. What can you conclude about the direction of the particle's velocity relative to the field? Give two situations in which this would occur.
4. The magnetic force on a moving charge is always perpendicular to the velocity. What is the consequence of this for the speed, kinetic energy, and direction of the charge? Compare this to the effect of an electric force.
5. Two parallel wires carry currents in the same direction. Do they attract or repel? Explain using the right-hand rule for the field of one wire and the force on the other.
6. Explain why an electric motor would not continue rotating without a commutator (or equivalent current-reversal mechanism). What problem does the commutator solve?
7. A compass placed on a table near a straight wire carrying a large current is deflected from pointing north. Describe the direction of the deflection and explain why it occurs.
8. Explain in terms of atomic structure why heating a permanent magnet above its Curie temperature destroys the magnetism. What does temperature do to the magnetic domains?

Magnetic Field Diagram Exercises

9. Sketch the magnetic field lines (a) around a bar magnet (showing at least 6 lines, including at least two that show the internal field); (b) between two north poles brought near each other; (c) between a north and south pole brought near each other. Label all poles and indicate field direction with arrows on each line.
10. A long straight wire runs left to right across your page, carrying conventional current flowing to the right. Sketch the magnetic field lines around the wire at four positions: directly above, below, to the left, and to the right of the wire. Indicate the direction of B at each position using the right-hand rule.
11. Sketch the field pattern of a solenoid with 8 turns, indicating the north and south poles of the solenoid field and the direction of conventional current flow in the coil. How would adding a soft iron core change the field? Describe without re-sketching.
12. Two parallel wires, A and B, are 4 cm apart and carry currents of 3 A (wire A) and 6 A (wire B) in the same direction. Sketch (roughly to scale) the positions of the wires and indicate where the net magnetic field is zero between them. (Hint: the zero point is where the fields from each wire are equal and opposite. The field from each wire at distance r is $B = \mu_0 I/2\pi r$.)

Magnetic Force Problems

13. An electron (charge $q = -1.6 \times 10^{-19}$ C, mass $m = 9.11 \times 10^{-31}$ kg) moves at 5.0×10^{6} m/s perpendicular to a magnetic field of 0.15 T. Find (a) the magnetic force magnitude, (b) the radius of the circular path, and (c) the period of revolution.
14. A proton moves with velocity 2.0×10^{5} m/s at an angle of 30° to a magnetic field of 0.50 T. Find the force on the proton. ($q_{proton} = 1.6 \times 10^{-19}$ C)
15. A horizontal wire of length 0.25 m carries 8.0 A of current directed east. It sits in a magnetic field of 0.050 T directed vertically upward. Find the magnitude and direction of the force on the wire.
16. A charged particle of mass 2.0×10^{-26} kg and charge 3.2×10^{-19} C moves in a circle of radius 0.080 m in a field of 0.40 T. Find (a) the speed of the particle, (b) the magnetic force, and (c) the period of revolution.
17. Two particles with the same speed and charge enter the same magnetic field perpendicular to the field lines. Particle A has mass m; Particle B has mass 4m. Compare their orbital radii and periods of revolution.

Electromagnet Applications

18. A solenoid has 500 turns, is 0.20 m long, and carries a current of 2.0 A. Find the magnetic field inside the solenoid (a) without a core and (b) with a soft iron core of relative permeability $\mu_r = 2000$.
19. An electromagnet is used to lift a steel beam. The magnetic force must equal the weight of the beam (mass 800 kg). If the electromagnet has pole-face area 0.040 m^2, and the magnetic pressure (force per area) is $P = B^2/(2\mu_0)$, find the required B field to support the beam's weight.
20. A solenoid of 800 turns/m carries 5.0 A. Find the field inside. If the current is doubled and the length is doubled (keeping the same number of turns per metre), what happens to the field?
21. A relay contains a solenoid with 200 turns, core length 0.05 m, and core relative permeability 500. The relay is activated when $B \geq 0.020$ T inside the core. Find the minimum current needed to activate the relay.

Motor and Current Analysis

22. A rectangular coil in a motor has 50 turns, dimensions 8.0 cm × 12 cm, and carries 2.0 A in a field of 0.30 T. Find the maximum torque on the coil ($\tau = nBIA$ for a coil of n turns and area A, at maximum).
23. A DC motor runs at 1200 rpm on a 120 V supply and draws 3.5 A. Find (a) the electrical power input, (b) if the motor is 85% efficient, the mechanical power output, and (c) the torque at the output shaft.
24. A wire carrying 10 A runs east–west. It is located 5.0 cm north of a compass. The compass reads 60° east of magnetic north (which in this location equals geographic north). (a) Find the magnetic field from the wire at the compass location. (b) Is the compass deflected east or west? Explain.
25. Two parallel wires, each carrying 15 A in the same direction, are separated by 3.0 cm. Find the force per unit length between the wires. Is it attractive or repulsive? Show the calculation using $F/L = \mu_0 I_1 I_2/(2\pi d)$.

Chapter 44 Checkpoint

Before moving on to Chapter 45, confirm you can do each of the following.

I can...	See Section
Explain the origin of magnetism at the atomic and macroscopic level	44.1 / 44.3
State the law of magnetic poles and explain why monopoles do not exist	44.2
Describe magnetic domains and explain permanent and induced magnetisation	44.3
Describe magnetic field lines and the rules for drawing them	44.4
Explain Earth's magnetic field and define magnetic declination	44.5
Calculate the force on a moving charge: $F = qvB \sin\theta$; apply the right-hand rule	44.6
Calculate the radius of circular motion of a charged particle in B: $r = mv/qB$	44.6
Calculate the force on a current-carrying wire: $F = BIL \sin\theta$	44.7
Calculate the magnetic field near a straight wire: $B = \mu_0 I/(2\pi r)$	44.8
Describe how electromagnets and solenoids work; calculate $B = \mu_0 nI$	44.9
Explain the operating principle of a DC electric motor	44.10
Classify magnetic materials as ferromagnetic, paramagnetic, or diamagnetic	44.11
Identify and avoid common mistakes in magnetism problems	44.13

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